MANSI PATIL

Data Scientist | Machine Learning | Generative AI

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PROFESSIONAL SUMMARY

Computer Science graduate (B.Tech, CGPA 8.74/10) with hands-on experience in data science, machine learning, and Generative AI through an industry internship and end-to-end projects in fraud detection, loan risk, retail analytics, and demand forecasting. Skilled in SQL, Python, and PySpark for data preparation, predictive modeling, forecasting, and dashboards, with MLflow experiment tracking, FastAPI/Docker deployment, and exposure to NLP and RAG on unstructured data. Seeking a Data Scientist role to build reliable, data-driven solutions.

CORE SKILLS

Programming & Query Languages: Python, SQL, PySpark
Machine Learning: Scikit-learn, XGBoost, Statistical & Predictive Modeling, Regression, Classification, Clustering, Time-Series Forecasting, Feature Engineering, Model Evaluation, Validation & Cross-Validation
Deep Learning & NLP: PyTorch, Natural Language Processing (NLP), OpenCV, YOLO
Generative AI / LLM: LLM Integration, Prompt Engineering, RAG, Multimodal RAG, Embeddings, Vector Databases (Qdrant), Cross-Modal Retrieval, LangChain, LangGraph, OpenAI GPT, Claude, Gemini, Groq API, Hugging Face
MLOps & Deployment: MLflow, Evidently AI (Model Monitoring), Docker, CI/CD (GitHub Actions), Git/GitHub, FastAPI, REST APIs
Big Data & Data Engineering: PySpark, ETL Pipelines, Data Cleansing & Validation, Data Transformation, Pandas, NumPy
Analytics & Visualization: Power BI, Streamlit, Plotly, Matplotlib/Seaborn, Exploratory Data Analysis, Advanced Excel
Backend, Databases & Cloud: Django, MySQL, PostgreSQL, SQLite, Supabase, AWS, Postman, Jupyter

EXPERIENCE

Data Science Intern | Amdox Technologies (GitHub) *Jan 2026 – Jun 2026*

- Developed 3 ML models for NeuralRetail, an enterprise retail analytics system, in collaboration with senior team members — XGBoost churn classification (0.84 ROC-AUC), ARIMA 30-day demand forecasting (12.3% MAPE), and K-Means customer segmentation.
- Built and tested secure RESTful FastAPI endpoints with JWT authentication and SQLite persistence, powering real-time model scoring and automated inventory reorder alerts.
- Performed exploratory data analysis, data cleaning, and validation, engineered RFM and lag features, and built an interactive Streamlit/Plotly dashboard to communicate sales trends and churn risk to stakeholders.
- Tracked all 3 models using MLflow, added Evidently AI drift reports for model monitoring, and containerized services with Docker.

PROJECTS

Real-Time UPI-Style Transaction Fraud Detection System (GitHub)
Python, XGBoost, FastAPI, Streamlit, PySpark, Scikit-learn

- Engineered an end-to-end fraud detection pipeline on 284,807 transactions (European credit card benchmark) using an XGBoost Classifier, handling severe class imbalance (~0.17% fraud rate) with cost-sensitive learning (scale_pos_weight=1000).
- Built data ingestion with PySpark and feature scaling with Scikit-learn (StandardScaler) to prepare the dataset for model training.
- Developed REST APIs using FastAPI and Pydantic validation to serve real-time model inference and probability-based risk tiers (LOW, MEDIUM, HIGH).
- Built an interactive Streamlit dashboard simulating live UPI transactions (amount, merchant type, location) with real-time risk scores, probability breakdown, and fraud alerts.

Multimodal RAG Engine — Cross-Modal Document Intelligence System (GitHub)
Python, FastAPI, Qdrant, CLIP, BLIP-2, Claude API, Docker

- Built a multimodal RAG pipeline extracting text, tables, and high-DPI page images from unstructured PDFs (PyMuPDF), with sentence-aware chunking (~300 characters) to eliminate mid-sentence breaks.
- Designed a dual-vector retrieval system combining 384-dimension SentenceTransformer text embeddings and 512-dimension OpenCLIP image embeddings via Reciprocal Rank Fusion, enabling joint retrieval of text and visual chart content.
- Reduced retrieved context by over 60% (to 3 high-precision chunks per query) by applying a dynamic relevance cutoff (≥0.40) and MS-MARCO Cross-Encoder reranking in Qdrant, cutting noise while preserving answer accuracy.
- Engineered a grounded Vision-LLM generator (Claude API) with XML-based prompt-injection defenses, inline citations, and refusal logic to prevent hallucinated answers, validated by an automated regression-testing suite; deployed via FastAPI with a Streamlit UI.

Loan Approval Prediction System with AI Chatbot (GitHub | Live Demo)
Python, Scikit-learn, XGBoost, FastAPI, Streamlit, OpenAI API (GPT-4o-mini)

- Delivered an end-to-end ML classification pipeline (XGBoost, Scikit-learn) to assess applicant financial risk and predict loan approval status based on 10+ demographic and credit variables, evaluated with cross-validation.
- Integrated an AI financial advisory chatbot (OpenAI GPT-4o-mini) to explain prediction rationale and provide actionable, data-grounded, and interpretable recommendations.
- Deployed an interactive Streamlit analytics dashboard alongside the full-stack system, taking it live to cloud production via Render.

EDUCATION

Maharashtra Institute of Technology — B.Tech, Computer Science Engineering | CGPA: 8.74 / 10 *Jan 2022 – May 2026*

CERTIFICATIONS

Business Intelligence with Advanced Excel and Power BI (2024) • CSRBOX — Agentic AI: From Learner to Builder – AI Agent Architect (2025) • Deloitte — Data Analyst Certificate (2025)